

WASP-52b

R-1209

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_201027 · created 2026-07-31T07:38:01+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459149.7523 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.211 +/- 0.063
Transit depth (Rp/Rs)^2	4.46 +/- 2.67 [%]
Orbital Inclination (inc)	85.67 +/- 2.76
Airmass coefficient 1 (a1)	0.7 +/- 0.42
Airmass coefficient 2 (a2)	0.24 +/- 0.54
Scatter in the residuals of the lightcurve fit is	61.01 %
Best Comparison Star	None
Optimal Aperture	1.34
Optimal Annulus	5.25
Transit Duration (day)	0.036 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.211 ± 0.063 vs 0.1646 (deviation 0.74σ)
Duration fit vs published	0.036 d vs 0.0758 d (deviation 3.32σ)
Mid-time O–C	-13.8 min
Depth SNR	3.3
Residual scatter	61.01 %

Light curve

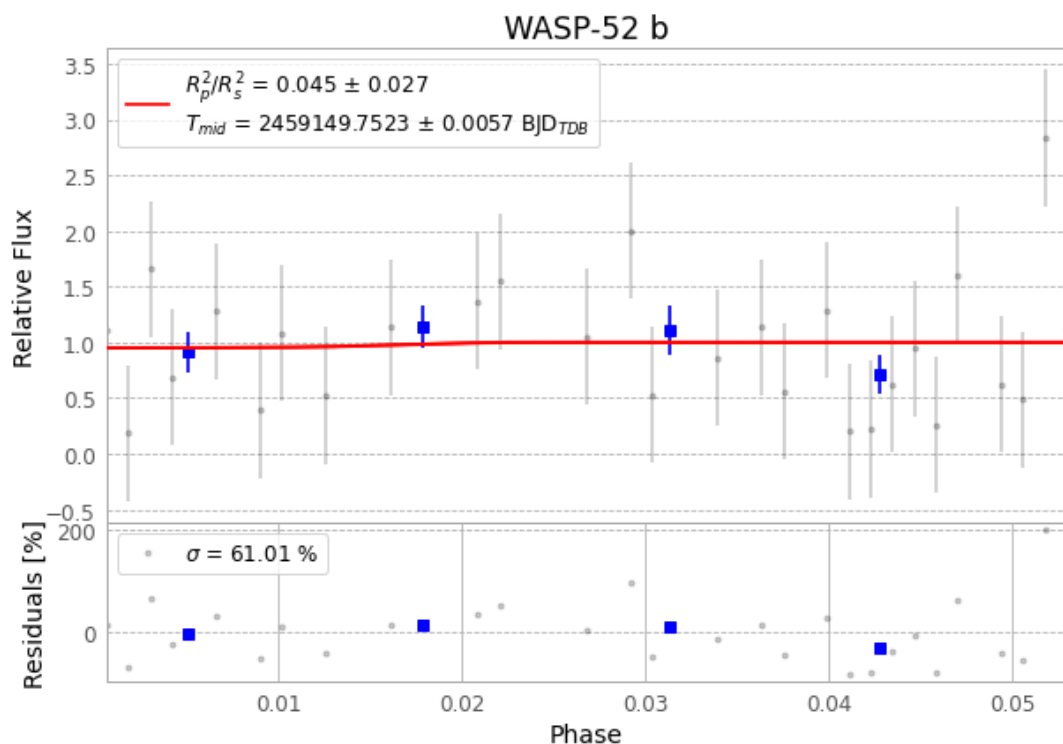


Figure 1 — FinalLightCurve_WASP-52 b_2020-10-26.png (SHA-256 6a7b2ab8a9492d01...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [579, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 93ac3e85962d1369...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

46 FITS frames (30 MB), WASP-52201027055412.FITS → WASP-52201027080913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-201027.log (857dd2641cee...), FinalLightCurve_WASP-52 b_2020-10-26.png (6a7b2ab8a949...), FinalLightCurve_WASP-52 b_2020-10-26.csv (d0477b7d06fb...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 07:38Z.